Chapter ~ 5

Magnetism & Matter.

=> Magnetic dipole moment of a current loop.

=> Magnetic dipole moment of a revolving electron.

Note: These both topics are same as Chapter 4 : Torque and Electron as magnetic dipole.

=> Bar Magnet

All bar magnets have two poles, the north pole and the south pole. It will always have it.

=> Cutting a Bar Magnet.

i] Along its length : 

Pole strength reduces by halves

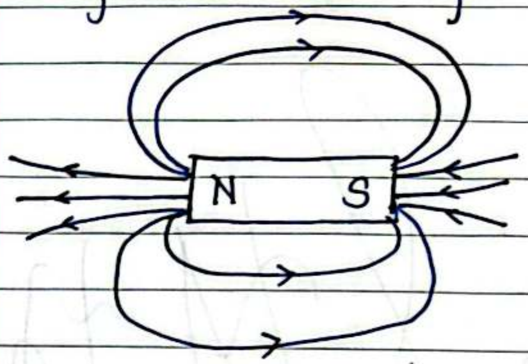
Length remains same.

ii] Perpendicular its length : 

Pole Strength remains same

Length reduces by half.

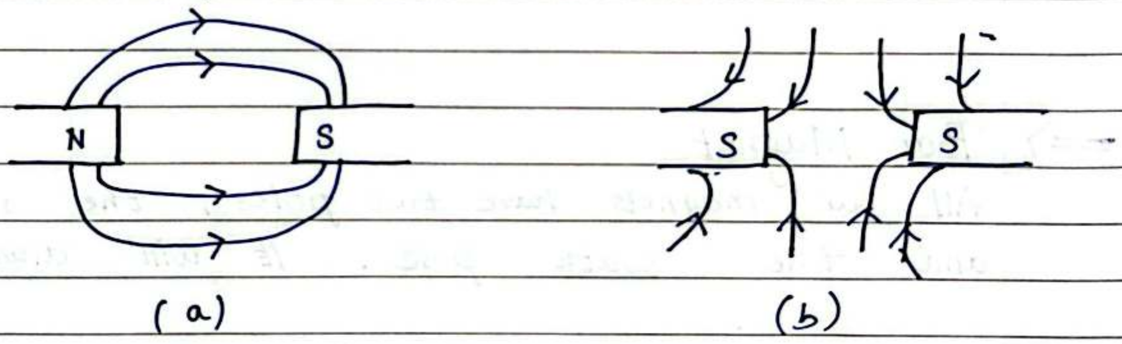
⇒ Magnetic Lines of Force.



the lines of force in a magnetic field are those imaginary lines which continuously represent the direction of magnetic field.

Properties.

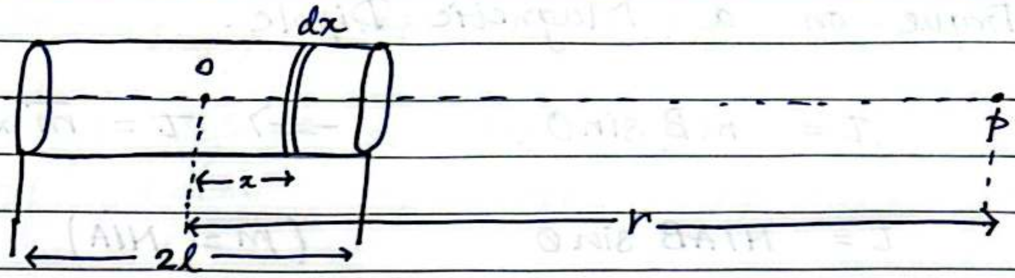
i] The magnetic lines emerge from north pole and following curved path reach south pole.
ii] Two lines of force never intersect each other.
iii] Near the magnetic poles where field is stronger the lines of force are closer.
iv) No lines of force will pass through neutral point.



⇒ A current carrying Solenoid : A Bar Magnet.

⟼ It always rests in the north-south direction.

⟶ It is just like a bar magnet having north and south poles.
⟶ two if placed will have attraction or repulsion.



⇒ Magnetic Field of a magnetic dipole / Small bar magnet.

i) End on Position :

$$dB = \frac{\mu_0}{4\pi} \frac{2\pi (n dx) I a^2}{[a^2 + (r-x)^2]^{3/2}}$$

$$\int dB = \frac{\mu_0}{4\pi} 2\pi n I a^2 \int_{-l}^{l} \frac{dx}{[a^2 + (r-x)^2]^{3/2}}$$

$$= \frac{\mu_0}{4\pi} 2\pi n I a^2 \int_{-l}^{l} \frac{-x' dx}{r^3}$$

$$= \frac{\mu_0}{4\pi} \frac{2\pi n I a^2}{r^3} (l + l)$$

$$= \frac{\mu_0}{4\pi} \frac{2\pi n I a^2}{r^3} \times 2l$$

$m = \pi N I a^2 \times 2l$

$$\boxed{B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}}$$

ii) Broad side on Position : $(l <<< r)$

$$\boxed{B = \frac{\mu_0}{4\pi} \frac{m}{r^3}}$$

Note : $B_{end} = 2 \times B_{broad}$.

⇒ Torque on a Magnetic Dipole.

$\tau = mB\sin\theta$. ⇒ $\vec{\tau} = \vec{m} \times \vec{B}$

$\tau = NIAB\sin\theta$ $(m = NIA)$.

Diamagnetic Substances	Paramagnetic Substances	Ferromagnetic Substances
⇒ feebly magnetised opp to direction of magnetic field	feebly magnetised in the direction of magnetic field	strongly magnetised in the direction of magnetic field.
⇒ they are repelled away from a magnet.	they are attracted towards a magnet.	they are attracted fast towards a magnet.
⇒ Examples : Bismuth zinc, copper, silver lead, water, mercury NaCl, nitrogen, hydrogen	Aluminium, sodium platinum, manganese antimony, $CuCl_2$ liquid oxygen	Iron, Nickel Cobalt, magnetite (Fe_3O_4) etc.

As a matter of fact : every substance is in general diamagnetic.

⇒ Some Important Terms used in Magnetism

i) Magnetic Induction (B) : when any substance is placed in an external magnetic field the magnetism produced is induced magnetism.

The number of magnetic lines of induction inside a magnetised substance crossing unit area is called magnitude of magnetic Induction.

ii] Intensity of Magnetisation (M)

It is defined as the magnetic moment per unit volume of the magnetised substance.

$$M = \frac{m}{V}$$ $m \longrightarrow$ magnetic moment.
$V \longrightarrow$ volume.

iii] Magnetic Intesity / Magnetic Field Strength (H).

$$H = \frac{B}{\mu_0} - M.$$

It is the capability of magnetising field to magnetise the substance.

iv] Magnetic Permeability (μ)

It is defined as the ratio of magnetic Induction B inside the magnetised Substance to the magnetic Intensity (H)

$$\mu = \frac{B}{H.}$$

v] Relative Magnetic Permeability (μ_r)

$$\mu_r = \frac{\mu}{\mu_0}$$ $\mu \longrightarrow$ magnetic permeability
$\mu_0 \longrightarrow$ magnetic permeability in free space.

$$\mu_r = \frac{B}{B_0}.$$

vi] Magnetic Susceptibility (χ_m)

It is defined as the ratio of the intensity of magnetisation to the magnetic Intensity of magnetising field.

$$\chi_m = \frac{M}{H}.$$

χ_m of Diamagnetic $\Rightarrow$ small and negative ($\chi_m < 0$)

χ_m of Paramagnetic $\Rightarrow$ small positive values ($\chi_m > 0$)

χ_m of Ferromagnetic $\Rightarrow$ very large +ve ($\chi_m >> 0$).

$\Rightarrow$ Relation between relative permeability (μ_r) and Magnetic Susceptibility (χ_m).
Imp.

$$H = \frac{B}{\mu_0} - M$$

$$H + M = \frac{B}{\mu_0}$$

$$\mu_0 (H+M) = B.$$

$$\mu_0 (H+M) = \mu H$$

$$\mu_0 \left(1 + \frac{M}{H}\right) = \mu.$$

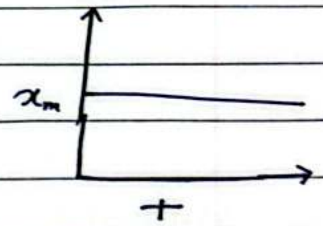
i] $\boxed{\mu_0 (1 + \chi_m) = \mu.}$

$$(1 + \chi_m) = \frac{\mu}{\mu_0}$$

ii] $\boxed{1 + \chi_m = \mu_r}$

⟹ Effect of Temperature on Magnetic Materials.

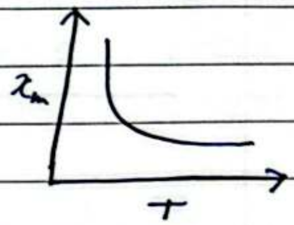
1) Diamagnetic Materials : It is independent of kelvin Temperature.



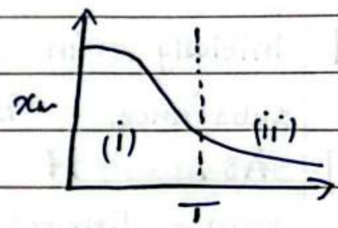
2) Paramagnetic Materials : It is inversely proportional to temp.

$$\chi_m = \frac{C}{\mu_0 T}$$

Curies Constant.



3) Ferromagnetic Materials : Temperature at which ferromagnetic material behaves as a paramagnetic material is called Curie's Temperature.



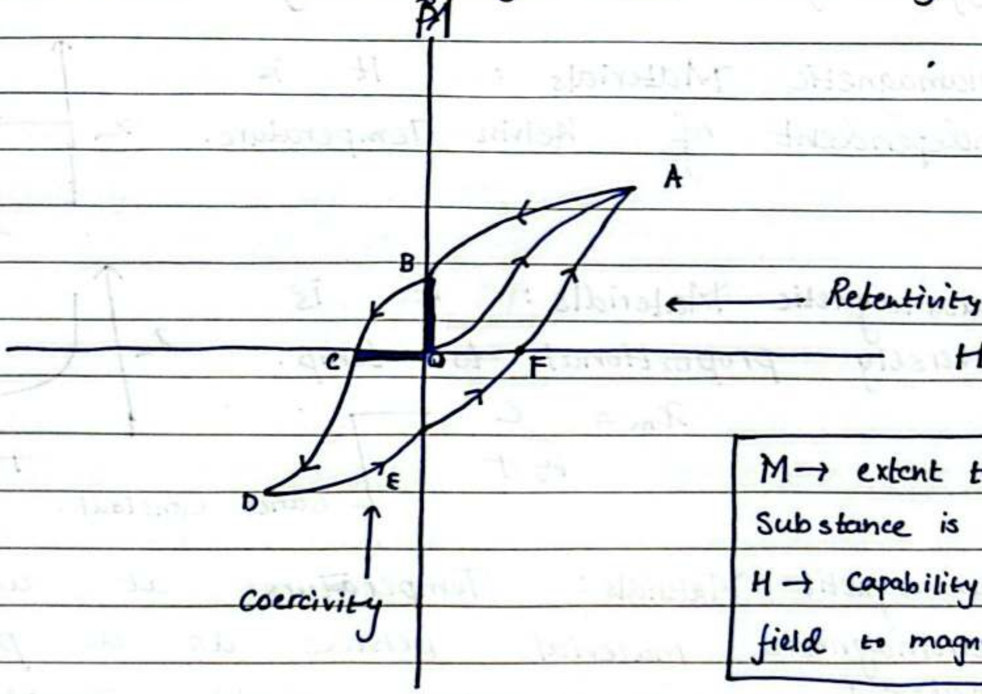
⟹ Curies Laws

It states that $\chi_m \propto 1/T$

magnetic susceptibility of a paramagnetic material is inversely proportional to the Temperature.

$$\chi_m \propto \frac{1}{T}$$

⇒ Hysteresis : Retentivity and Coercivity.



M → extent to which substance is magnetised
H → Capability of external field to magnetise substance.

i] Initially, in Absence of Magnetic Field the substance is in unmagnetised state.

ii] As, H is increased, M also increases non-linearly (point A is state of magnetic saturation of the substance).

iii] Any further increase in H does not produce any increase in M.

iv] If now the magnetising field H is decreased M also decreases along path AB, thus M lags behind.
When H becomes 0 (at point B)
M has still value OB.

v] The magnetisation remaining in the substance (when M is reduced to 0) is called residual Magnetism.

Resistivity – It is the measure of magnetisation remained in the substance when $H=0$

vi] If now H is increased in the reverse direction, M decreases along BC until it becomes 0 at C
(At point C $[M=0, H\neq0]$)
Point C $\Rightarrow$ Coercivity.

vii] $\Rightarrow$ Coercivity : It is the measure of the reverse magnetising field required to destroy residual magnetism of substance.

vii] As H is increased beyond OC the substance is magnetised in the opp. direction along CD. at D substance is again saturately magnetised.

viii] By taking H from its negative value to its original maxim value of symmetrical curve DEFA.

ix] At points B and E substance is magne-tised in absence of magnetic field.

The cycle of ABCDEFA is called cycle of magnetisation.

$$\boxed{\chi_m = \frac{H}{M}}$$

Hysteresis loss : Area under loop gives hysteresis loss.

The energy lost per unit volume of a substance in a complete cycle of magnetisation which is equal to the area of hysteresis loop is called hysteresis loss.

⇒ Difference in magnetic properties of soft iron and steel.

retentivity soft iron > retentivity of steel

coercivity soft iron < coercivity of steel.

∴ Hysteresis loss in soft iron is smaller than that in steel.

Note: Material with minimum loss is most suitable.

⇒ Selection of magnetic materials

i] Permanent magnets : high retentivity
high coercivity.

ii] Electromagnets : Low retentivity
(soft iron).

iii] Transformers : Low Hysteresis Loss.